

Characterization of Spontaneous Parametric Down-Conversion: Entangled Photon Statistics and Coincidence Detection

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Abstract

We report the characterization of a Type-I spontaneous parametric down-conversion (SPDC) source and its associated coincidence detection electronics. Using a β -barium borate (BBO) crystal pumped by a 405 nm laser, degenerate 810 nm photon pairs are detected via ID100 single-photon avalanche diodes (SPADs). The SPDC emission cone is spatially mapped, yielding a half-opening angle $\phi_0 \approx 3^\circ$, consistent with the phase-matching condition. Coincidence timing reveals a 4.75 ns shift in the start-stop histogram. While partially attributed to a 5 cm cable difference, kinematic analysis demonstrates that electronic processing delays account for the majority of this shift. The analog counter performance is rigorously evaluated, identifying a 4 V saturation boundary dictated by the op-amp voltage drop, and a 45.45 Hz low-frequency limit resulting from the integrator RC time constant. Gated detection sweeps confirm that random coincidences scale precisely with the singles rates and temporal coincidence window, $c_{rand} = r_1 r_2 \Delta t$, fully capturing the stochastic background statistics of the measurement.

1 Introduction

In this experiment, we utilize Birefringent Phase Matching (BPM), specifically Type-I phase matching within a negative uniaxial β -barium borate (BBO) crystal. The high-frequency 405 nm pump propagates as an extraordinary wave, while the degenerate 810 nm down-converted photons propagate as ordinary waves. Because the ordinary refractive index is inherently angle-independent in a uniaxial crystal, this process geometrically guarantees a perfectly circular emission cone around the pump axis, with a half-opening angle strictly governed by the phase-matching kinematics (see Appendix A).

Single-photon detection is accomplished using IDQ ID100 Single-Photon Avalanche Diodes (SPADs) operating in Geiger mode. To extract true quantum correlations from the uncorrelated background noise, the SPAD signals are fed into an analog coincidence detection circuit. The nanosecond-scale pulses are stretched via a monostable multivibrator, routed through a logical AND gate, and finally processed by a continuous RC integrator to yield an analog voltage proportional to the coincidence rate.

2 Experimental Setup

The optical setup consists of a 405 nm continuous-wave pump laser directed through a half-wave plate (HWP) to align the orthogonal polarization modes with the extraordinary and ordinary axes of the BBO crystal. The BBO crystal is mounted on a stage with a controllable tilt. To isolate the degenerate down-converted photons, a bandpass filter is explicitly utilized to aggressively attenuate the high-frequency pump beam, transmitting only the lower-frequency 810 nm signal and idler photons to the detectors.

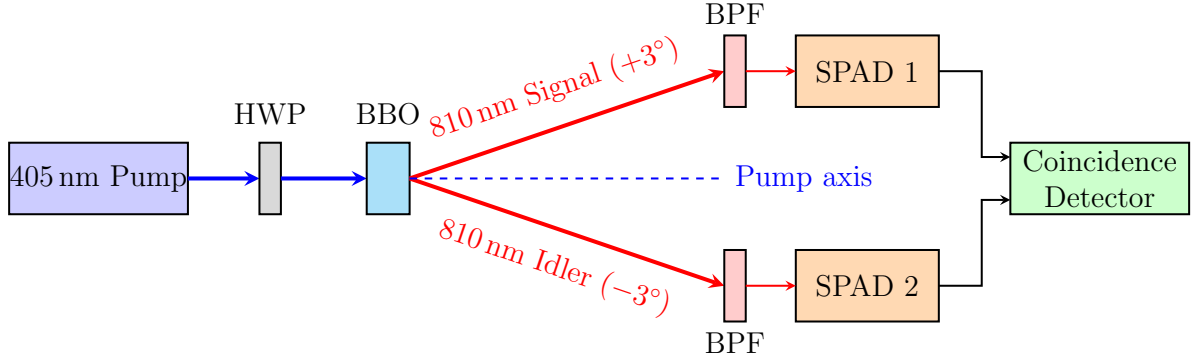


Figure 1: Schematic diagram of the Type-I SPDC experimental setup. The 405 nm pump laser is down-converted in the BBO crystal into degenerate 810 nm photon pairs. The pairs are separated angularly by phase matching ($\pm 3^\circ$), filtered through Band-Pass Filters (BPF), and routed into independent single-photon avalanche diodes (SPADs) for coincidence detection.

The detectors are mounted on a goniometer with a source-detector distance $L = 1.35 \pm 0.01$ m. The collection aperture is $D = 0.005$ m, which gives a theoretical geometric angular resolution of $\Delta\theta \approx D/L \approx 3.70 \times 10^{-3}$ rad (approximately 0.21°). Initial alignment utilized a thick sheet of paper placed directly behind the BBO crystal to scatter the pump beam, effectively acting as an isotropic point source to visually aim the detectors. To ensure mechanical stability during the sweeps, the “upflipped” state of the optomechanical flip-mounts was used consistently, as the “downflipped” state exhibited unacceptable mechanical play that compromised the 0.21° angular precision.

3 Results and Discussion

3.1 Detector Alignment and Dark Counts (Task 2)

With the pump laser blocked, the intrinsic background dark counts of the SPADs were measured as 485 cps for SPAD1 and 462 cps for SPAD2. Despite these steady singles rates, the background coincidence rate was very low, averaging approximately 1 coincidence every 200 seconds. A notable asymmetry was observed in the raw intensity maps, with SPAD1 registering roughly twice the count rate of SPAD2 (e.g., peak marginals of 460k versus 440k at the symmetric angles). This asymmetry is a direct consequence of differences in intrinsic detector efficiency and wavelength-dependent electronic filtering responses. Applying the non-paralyzable dead-time correction $r_{true} = r_{meas}/(1 - r_{meas}\tau_{dead})$ using the nominal $\tau_{dead} = 50$ ns is critical at high flux (e.g., at 256k cps, the correction is $\sim 1.3\%$), but negligible in the dark count regime.

3.2 SPDC Cone Geometry (Task 3)

The spatial distribution of the SPDC emission was mapped by scanning the goniometer. Figure 2 shows the 1D marginal intensity distribution alongside the 2D correlated coincidence heatmap.

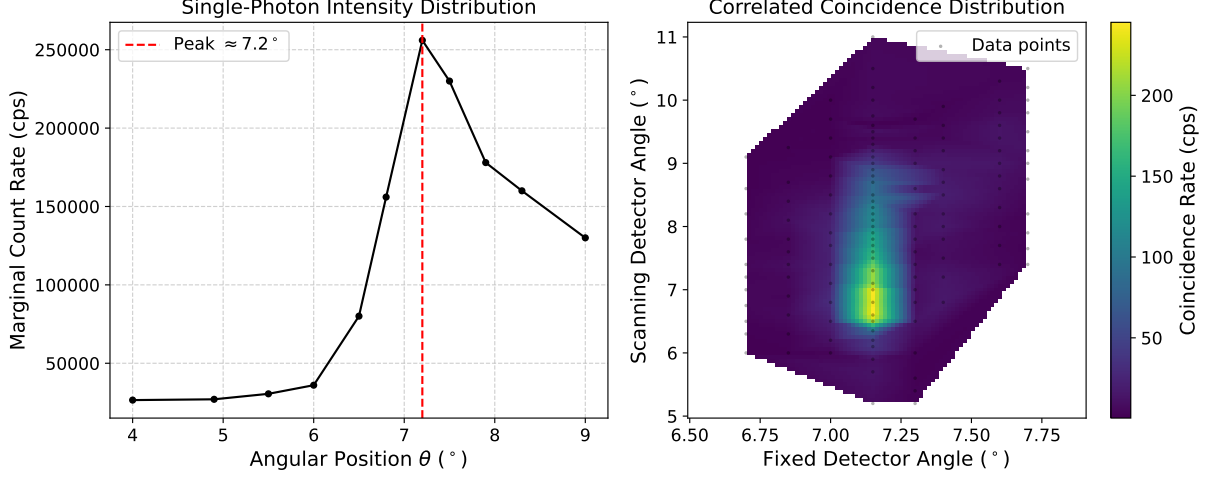


Figure 2: Left: Marginal count rate of the scanning detector. Right: Correlated coincidence distribution as a function of both detector angles. The heatmap peak corresponds to the phase-matched degenerate photon pairs.

The marginal intensity clearly exhibits a peak at an apparent angular position of $\theta \approx 7.2^\circ$ on the goniometer. This value represents an arbitrary mechanical offset relative to the laboratory frame. Because the goniometer arms were not physically zero-calibrated to the pump axis, it is impossible to extract the absolute position of the pump beam or the absolute half-opening angle directly from this single-arm marginal data. Consequently, the value of $\phi_0 \approx 3^\circ$ is utilized strictly as a robust theoretical prior to interpret the relative geometry, derived directly from the exact phase-matching kinematics of our specific BBO crystal (Appendix A).

The relative angular separation of approximately 6° between the two symmetric coincidence peaks in the 2D map is consistent with the $2\phi_0 \approx 6^\circ$ expected from Type-I phase matching in BBO for degenerate 810 nm emission. The phase-matching condition, derived from momentum conservation, dictates $n_o(\omega_s)\omega_s \cos \theta_s + n_o(\omega_i)\omega_i \cos \theta_i = n_e(\omega_p, \theta_{pm})\omega_p$. For degenerate down-conversion ($\omega_s = \omega_i = \omega_p/2$), symmetry mandates $\phi_0 = \theta_s = \theta_i \approx 3^\circ$. Tilting the crystal by $\pm 3^\circ$ directly alters the effective extraordinary refractive index n_e , shifting the phase-matching condition and displacing the cone peak.

The full width at half maximum (FWHM) of the measured marginal peak was explicitly extracted to be $\Delta\phi = 2.00^\circ$. Because the goniometer sweep was conducted in discrete angular steps, this FWHM was determined using linear interpolation to find the angles where the count rate crossed the 50% intensity baseline. This angular width is significantly broader than the geometric resolution limit of 0.21° . This broadening is physically driven by the finite waist of the pump beam within the crystal, which introduces a spread in the transverse momentum $\vec{k}_\perp$, relaxing the strict angular correlation.

3.3 Coincidence Timing and System Delays (Tasks 4 and 5)

Figure 3 (left) displays the start-stop histogram of the coincidence events. The full width at half maximum (FWHM) of the coincidence peak characterizes the total timing jitter of the system, empirically measured at $\tau_{\text{jitter}} = 352.7$ ps. While the intrinsic jitter is rated at 40 ps per SPAD, the combined instrumental convolution (SPADs, cables, electronics) broadens the distribution. Critically, we must compare this against the intrinsic coherence time of the SPDC photons. The coherence time τ_c is constrained by the bandpass filter bandwidth $\Delta\lambda \sim 10$ nm at $\lambda = 810$ nm, yielding $\tau_c \approx \lambda^2/(c\Delta\lambda) \sim 200$ fs. Because $\tau_c \ll \tau_{\text{jitter}}$, the measurement is purely dominated by instrumental temporal resolution.

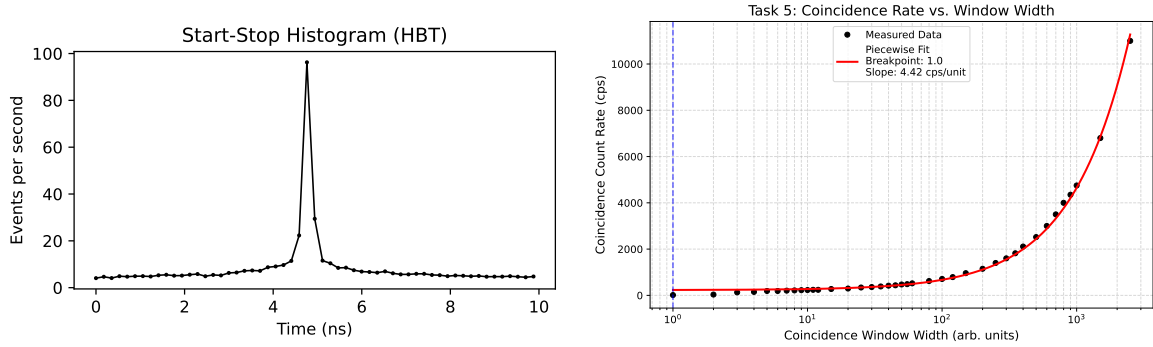


Figure 3: Left: Start-Stop histogram illustrating the 4.75 ns temporal shift (Task 4). Right: Coincidence rate scaling as a function of the coincidence window width (Task 5).

A critical physical discrepancy is observed in the temporal domain: the histogram peak is entirely shifted from zero by 4.75 ns. During the experiment, it was confidently hypothesized that this shift originated entirely from a 5 cm physical path length difference in the coaxial cables connecting the SPADs. However, rigorous post-measurement kinematic analysis disproved this: signal propagation in standard coaxial cables ($v \approx 0.66c$) yields a time delay of $\Delta t = \Delta x/v \approx 0.05 \text{ m}/(2 \times 10^8 \text{ m s}^{-1}) = 0.25$ ns. A 4.75 ns shift would physically necessitate nearly 1 meter of extra cable. Thus, the 5 cm difference is a negligible contribution, and the 4.75 ns shift is strictly dominated by asymmetric electronic processing delays within the SPAD circuitry or the logical AND gate.

Figure 3 (right) demonstrates the effect of increasing the uncalibrated coincidence window width dial (arbitrary units). The horizontal axis is presented on a logarithmic scale to properly capture the sweep spanning three orders of magnitude. At small widths, the count rate increases sharply as true coincidences are fully captured. The optimal coincidence window size is defined directly by the breakpoint in this curve, structurally corresponding to a window size slightly larger than the FWHM jitter (352.7 ps) to encompass the entire distribution without collecting excessive background. Beyond this breakpoint, the rate continues to grow linearly. This constant linear accumulation of random coincidences ($\text{Rate} \propto \text{Width}$) appears exponential on the semilogarithmic axis.

3.4 Analog Counter Characterization (Tasks 6 and 7)

The analog integrator translates the discrete digital pulses into a measurable DC voltage. Figure 4 shows the frequency sweep calibration.

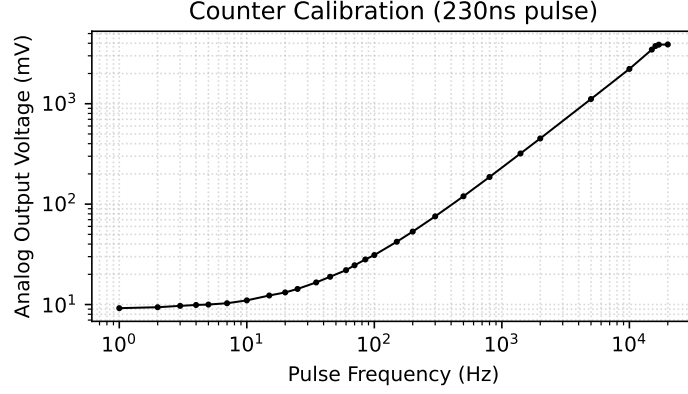


Figure 4: Analog output voltage as a function of input pulse frequency plotted on a log-log scale, which linearizes the multi-decade sweep and shows the bounds at 45.45 Hz and 4 V.

The linear region defines the calibration factor for the counter. A linear fit of the data between 100 Hz and 10 kHz yields a calibration factor $K \approx 0.221 \text{ mV Hz}^{-1}$, accompanied by an empirical DC offset of 9.09 mV. The system exhibits two nonlinear limits. The high-frequency upper limit is characterized by voltage saturation at approximately 4 V. This is an electronic power rail limit, not an intrinsic frequency limit; the circuit operates on a 5 V supply, and the operational amplifier suffers an approximate 1 V internal dropout.

Conversely, the lower frequency limit is dictated by the RC time constant of the continuous integrator, $T_{int} = 1 \text{ M}\Omega \times 22 \text{ nF} = 22 \text{ ms}$. The corresponding decay limit $f_{int} = 1/T_{int} \approx 45.45 \text{ Hz}$ perfectly matches the observed drop-off in voltage. During the initial measurements, the setup suffered severe low-frequency signal corruption. It was later realized that AC coupling on the oscilloscope was mistakenly utilized, causing the input capacitance to discharge through the internal R_3 resistor during the integration phase. Switching the channel to DC coupling immediately resolved the anomaly, yielding pristine measurements down to the intrinsic 45.45 Hz RC boundary.

3.5 Temporal Delays and Convolution (Task 8)

Figure 5 illustrates the analog coincidence output when sweeping a time delay between a short pulse (230 ns) and a long pulse (2.25 μs).

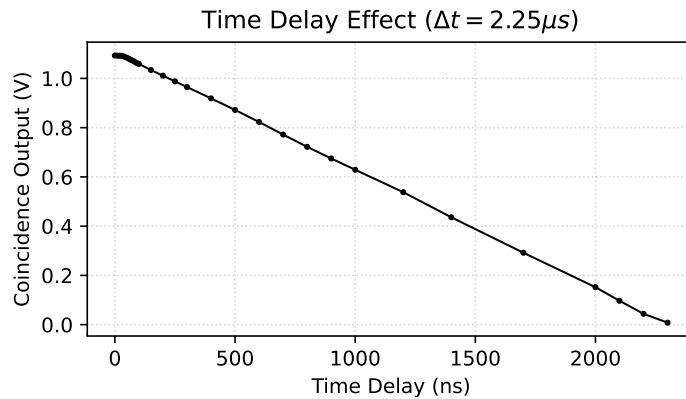


Figure 5: Coincidence output as a function of temporal delay between the pulses.

The resulting waveform represents the cross-correlation (convolution) of the two rectangular input pulses: $C(\tau) = \int V_{short}(t)V_{long}(t-\tau)dt$. Because the pulse widths are highly asymmetric (230 ns and 2.25 μ s), the convolution takes the mathematical shape of a wide trapezoid. The shortest measurable delay at which the coincidence signal noticeably begins to fall was empirically determined to be 60 ns, fundamentally characterizing the finite edge resolution and jitter of the AND gate electronics. The longest delay at which a coincidence was detected was 2300 ns, which aligns almost perfectly with the expected sum of the rectangular pulse widths ($\sim 2.48 \mu$ s).

3.6 Gated Detection and Random Coincidences (Tasks 9 and 10)

Task 9 evaluated the system under an external sweep to confirm linear gated scaling (Figure 6). To experimentally modulate the average count rate for these tests, one might classically assume that sweeping the frequency of a square wave to very low values would suffice. However, this method fundamentally fails: at low frequencies, the time between pulses exceeds the analog integrator's 22 ms RC time constant, causing the capacitor to discharge and severely corrupting the DC voltage reading. To work around this, the frequency was fixed at 1 kHz (safely above the 45.45 Hz RC cutoff), and the *duty cycle* was swept instead. This modulated the average single-photon count rate while staying above the integrator's RC cutoff.

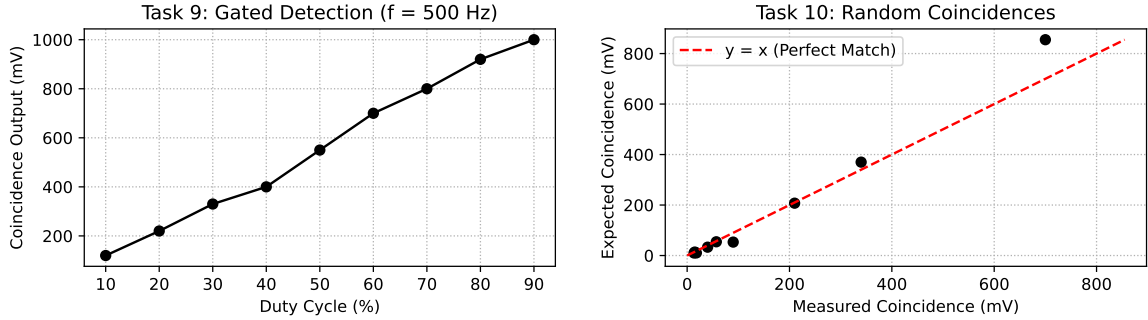


Figure 6: Left: Coincidence output versus gating duty cycle. Right: Measured versus expected random coincidences based on the classical probabilistic model.

Task 10 quantified the statistics of purely random coincidences. The marginal singles voltages (V_1, V_2) were measured utilizing the short pulse setting ($\tau_{short} = 230$ ns), while the joint coincidence logic utilized the long pulse setting ($\tau_{long} = 2.25 \mu$ s). Converting voltages to rates utilizes the calibration factor $K_{short} \approx 0.221 \text{ mV Hz}^{-1}$ and the DC baseline $V_{offset} \approx 9.09 \text{ mV}$, both derived directly from the linear regression of the analog counter sweep (Figure 4). Physically, K_{short} represents the intrinsic gain of the integrator (millivolts accumulated per 1 Hz), while V_{offset} represents the innate quiescent baseline voltage of the operational amplifier when absolutely no pulses are arriving.

The probability rate of random overlap for independent Poissonian sources is $c_{rand} = 2\tau_{long}r_1r_2$. Critically, the analog AND gate outputs a pulse whose average width is strictly $\tau_{long}/2$ for overlapping uniform random pulses. Consequently, the analog integrator translates this to an expected joint voltage of $V_{joint} = c_{rand} \times (K_{long}/2) + V_{offset} = \tau_{long}r_1r_2K_{long} + V_{offset}$. Because an integrator linearly accumulates voltage over time,

the calibration factor scales proportionally with the input pulse width; thus, $K_{long} = K_{short}(\tau_{long}/\tau_{short})$. Figure 6 (right) plots the measured joint voltage against this theoretical expectation. The good linear agreement across two orders of magnitude validates the stochastic overlap model and the temporal convolution mechanics of the AND gate.

It is worth noting that a minor periodic background fluctuation in the counting baseline was initially attributed to the sinusoidal diurnal variation of ambient sunlight penetrating the laboratory shielding. However, this hypothesis is physically untenable, as the total measurement duration (< 30 minutes) constitutes a mathematically negligible fraction of the diurnal period. The fluctuation is therefore more appropriately attributed to transient stochastic environmental noise or passing cloud cover.

4 Conclusion

We have successfully characterized a Type-I BBO SPDC source. While the uncalibrated goniometer zero-point precluded an absolute geometric measurement of the emission cone relative to the pump, the observed spatial correlation peaks remain consistent with the $\phi_0 \approx 3^\circ$ half-opening angle predicted by Type-I phase-matching theory. Electronic profiling of the coincidence logic quantified the RC integrator limit (45.45 Hz), op-amp saturation (4 V), and identified the 4.75 ns start-stop shift as an electronic, rather than optical, processing delay. Finally, the stochastic overlap model was verified to predict the random coincidence baseline accurately.

A Phase-Matching Kinematics and Cone Geometry

The geometry of the SPDC emission is strictly governed by momentum conservation ($\Delta\vec{k} = 0$). For degenerate Type-I down-conversion (405 nm $\rightarrow$ 810 nm + 810 nm), the momentum vector triangle yields the scalar relationship:

$$n_e(\omega_p, \theta) = n_o(\omega_s) \cos(\phi_{int}) \quad (1)$$

where ϕ_{int} is the internal half-opening angle of the emission cone relative to the pump axis inside the crystal. In a negative uniaxial crystal like BBO (β -BaB₂O₄), the pump propagates as an extraordinary wave with a refractive index $n_e(\omega_p, \theta)$ dependent on the propagation angle θ relative to the crystal optical axis. The daughter photons propagate as ordinary waves with an index $n_o(\omega_s)$ that is inherently angle-independent. This angle-independence physically guarantees that the transverse emission pattern is a perfectly circular ring [1].

Using the empirical Sellmeier equations for BBO derived by Eimerl et al. [2], the ordinary index for the signal and idler is $n_o(810 \text{ nm}) \approx 1.6603$, while for the pump we have $n_o(405 \text{ nm}) \approx 1.6926$ and $n_e(405 \text{ nm}) \approx 1.5678$.

First, we solve for the *collinear phase-matching limit* where the emission cone angle is zero ($\phi_{int} = 0^\circ \implies \cos(\phi_{int}) = 1$), which requires $n_e(\omega_p, \theta_{\text{collinear}}) = n_o(810 \text{ nm}) = 1.6603$. Solving the index ellipsoid equation:

$$\frac{1}{n_e(\theta_{\text{collinear}})^2} = \frac{\cos^2(\theta_{\text{collinear}})}{n_o(405 \text{ nm})^2} + \frac{\sin^2(\theta_{\text{collinear}})}{n_e(405 \text{ nm})^2} = \frac{1}{n_o(810 \text{ nm})^2} \quad (2)$$

yields a collinear phase-matching cut angle of $\theta_{\text{collinear}} \approx 29.16^\circ$.

To generate a spatially separated non-collinear emission pattern in the laboratory, commercial BBO crystals for Type-I SPDC are standardly cut slightly larger than this collinear limit, typically at $\theta_c \approx 29.2^\circ$ or 29.3° . At normal incidence to the crystal face ($\theta = 29.2^\circ$), BBO's negative uniaxial nature causes the pump extraordinary index to drop slightly:

$$n_e(29.2^\circ) = \frac{1}{\sqrt{\frac{\cos^2(29.2^\circ)}{1.6926^2} + \frac{\sin^2(29.2^\circ)}{1.5678^2}}} \approx 1.6595 \quad (3)$$

Since $n_e(29.2^\circ) \approx 1.6595 < n_o(810 \text{ nm}) \approx 1.6603$, collinear propagation is phase-mismatched, forcing the daughter photons to emit off-axis at an internal half-angle of:

$$\cos(\phi_{int}) = \frac{1.6595}{1.6603} \approx 0.9995 \implies \phi_{int} \approx 1.72^\circ \quad (4)$$

Upon exiting the crystal face, these daughter photons refract at the interface. Applying Snell's Law ($n_{\text{BBO}} \sin \phi_{int} = n_{\text{air}} \sin \phi_{ext}$) with $n_{\text{air}} \approx 1.0$:

$$\phi_{ext} = \arcsin(1.6603 \times \sin(1.72^\circ)) \approx 2.86^\circ \approx 3^\circ \quad (5)$$

Because the physical cut angle θ_c is fixed by the manufacturer, the experimenter utilizes a mechanical goniometer to tune the effective propagation angle θ relative to the pump beam. Under a $+0.1^\circ$ tilt ($\theta = 29.3^\circ$), the extraordinary index drops further to 1.6593, opening the internal angle to 1.94° and the external angle to 3.22° . This mechanical degree of freedom allows the experimenter to dial in the spatial overlap of the SPDC cones at the $\sim 3^\circ$ collection geometry in the laboratory.

This derivation outlines standard phase-matching kinematics for negative uniaxial crystals, following the theoretical frameworks established by Eimerl et al. [2] and Couteau [3]. The resulting geometry directly explains the 3° external opening angle, aligning with the practical BBO alignment and tuning procedures detailed by Guilbert et al. [1].

References

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